

① Electric Current (I)

The rate of flow of electric charge through a conductor is called electric current.

$$I = \frac{dq}{dt} \quad \begin{cases} I = \text{Current (A)} \\ dq = \text{Charge (C)} \\ dt = \text{Time (s)} \end{cases}$$

SI unit of current is Ampere (A)

$$1 \text{ A} = 1 \text{ C/s}$$

③ Ohm's Law

At constant physical conditions (temperature, pressure, etc.), the current flowing through a conductor is directly proportional to the potential difference across its ends.

$$V \propto I \Rightarrow V = IR$$

V = Potential difference (V)

I = Current (A)

R = Resistance (Ω)

$$R = \frac{V}{I}$$

Ohmic conductor $\rightarrow$ Obeys Ohm's law

Non-ohmic conductor $\rightarrow$ Does not obey Ohm's law

⑤ Resistivity (ρ)

It is the resistance of a conductor of unit length and unit cross-sectional area.

SI unit $\rightarrow \Omega \text{ m}$

Material	Resistivity ($\Omega \text{ m}$)
Silver	1.59×10^{-8}
Copper	1.68×10^{-8}
Aluminium	2.82×10^{-8}
Iron	9.71×10^{-8}
Nichrome	1.10×10^{-6}

⑥ Effect of Temperature

For metallic conductors, resistance increases with increase in temperature.

$$R_t = R_0 [1 + \alpha (T - T_0)]$$

R_t = Resistance at temperature T

R_0 = Resistance at reference temperature T_0

α = Temperature coefficient of resistance ($^{\circ}\text{C}^{-1}$)

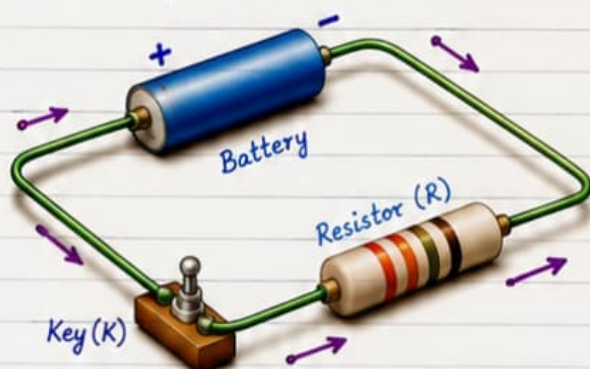
T = Temperature ($^{\circ}\text{C}$)



Key Point $\rightarrow$ Current electricity forms the basis of all electrical devices and circuits in daily life.

② Electric Circuit

A closed conducting path in which current flows is called an electric circuit.



Conventional current flows from positive terminal to negative terminal outside the battery.

④ Resistance (R)

Resistance is the property of a conductor to oppose the flow of current.

$$R = \rho \frac{l}{A}$$



ρ = Resistivity of material ($\Omega \text{ m}$)

l = Length of conductor (m)

A = Area of cross-section (m^2)

⑦ Combination of Resistors

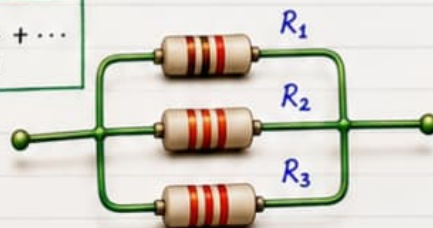
• Series Combination

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$



• Parallel Combination

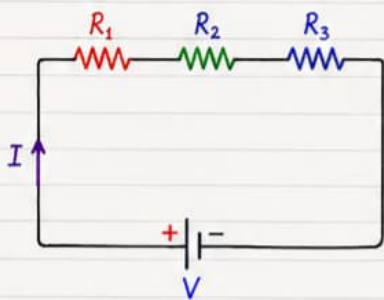
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$



8 Resistors in Series

- In series combination, resistors are connected end to end.
- Same current flows through each resistor.

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$



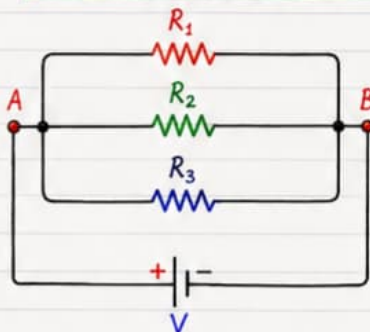
Key Points

- R_{eq} is greater than the greatest resistance.
- If one resistor breaks, the whole circuit stops working.

9 Resistors in Parallel

- In parallel combination, resistors are connected across the same two points.
- Same potential difference is across each resistor.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

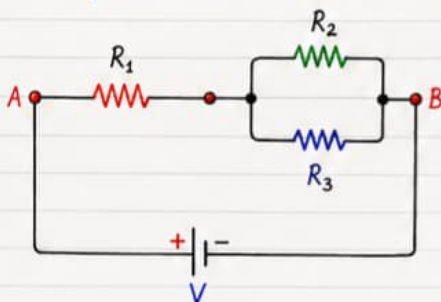


Key Points

- R_{eq} is smaller than the smallest resistance.
- If one resistor breaks, others continue to work.

10 Combination of Series and Parallel

Many circuits have both series and parallel parts.



How to solve?

- 1 Simplify the parallel part first.
- 2 Then combine in series.
- 3 Find total R_{eq} .
- 4 Use $I = \frac{V}{R_{eq}}$.

11 Electric Power (P)

The rate at which electrical energy is consumed in a circuit.

$$P = VI$$

Using Ohm's Law ($V = IR$)

$$P = I^2 R = \frac{V^2}{R}$$

Where,

- P = Power (W)
- V = Potential difference (V)
- I = Current (A)
- R = Resistance (Ω)

SI unit

Watt (W)
 $1 \text{ W} = 1 \text{ J/s}$



12 Joule's Law of Heating

The heat produced in a resistor is directly proportional to the square of current, resistance and time.

Where,

- H = Heat produced (Joule)
- I = Current (A)
- R = Resistance (Ω)
- t = Time (s)

$$H = I^2 R t$$

Applications

- Electric iron
- Geyser
- Heater
- Bulb



Example

Example 1:

Three resistors of 2Ω , 3Ω and 6Ω are connected in series with a 11 V battery. Find (i) current in the circuit (ii) potential difference across each resistor.

Solution:

$$R_{eq} = 2 + 3 + 6 = 11 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{11}{11} = 1 \text{ A}$$

$$V_1 = IR_1 = 1 \times 2 = 2 \text{ V}$$

$$V_2 = IR_2 = 1 \times 3 = 3 \text{ V}$$

$$V_3 = IR_3 = 1 \times 6 = 6 \text{ V}$$

Example

Example 2:

A 10Ω resistor is connected across a 12 V battery. Find (i) current (ii) power dissipated (iii) heat produced in 2 minutes.

Solution:

$$(i) I = \frac{V}{R} = \frac{12}{10} = 1.2 \text{ A}$$

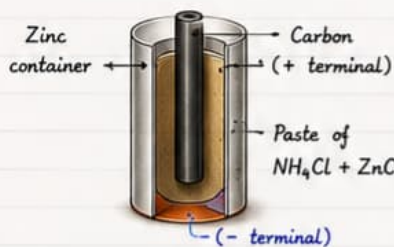
$$(ii) P = VI = 12 \times 1.2 = 14.4 \text{ W}$$

$$(iii) H = I^2 R t = (1.2)^2 \times 10 \times 120 = 1728 \text{ J}$$



13 Internal Resistance (r)

Every cell or battery has some resistance to the flow of current within itself. This is called its internal resistance.



Key Point

Internal resistance causes a drop in potential within the cell.

Terminal potential difference of a cell

$$V = E - Ir$$

Where,

E = emf of the cell

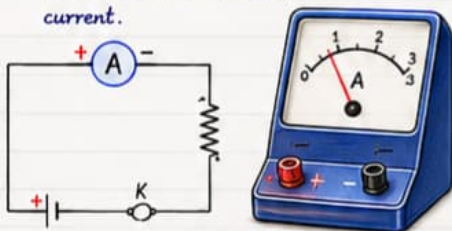
I = Current drawn

r = Internal resistance

15 Electric Measuring Instruments

(i) Ammeter

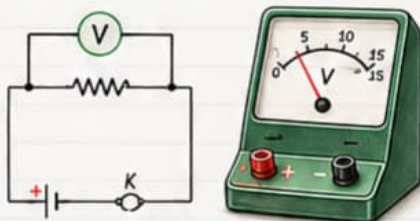
It is connected in series in the circuit to measure current.



- Very low resistance
- Connected in series
- Measures current (A)

(ii) Voltmeter

It is connected in parallel across the component.



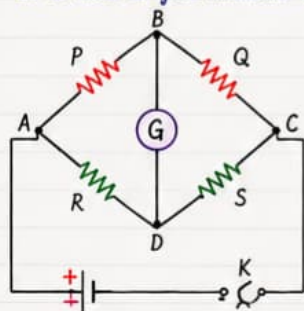
- Very high resistance
- Connected in parallel
- Measures potential difference (V)

Key Points

- Ammeter → series low resistance
- Voltmeter → parallel high resistance
- Never connect an ammeter in parallel or a voltmeter in series.

16 Wheatstone Bridge

It is used to find unknown resistance.



Balance Condition

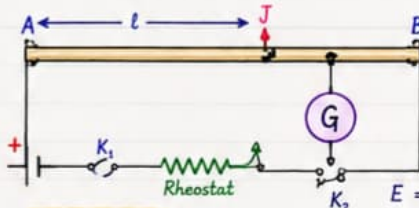
No current flows through galvanometer (G).

$$\frac{P}{Q} = \frac{R}{S} \text{ or } PS = QR$$

Used to measure an unknown resistance accurately.

17 Potentiometer

It is used to compare emfs of cells and to determine internal resistance.



$$\frac{E}{E'} = \frac{l}{L}$$

Key Point

At balance point, no current flows through galvanometer.

E = emf of cell
 E' = Known emf
 l = Balancing length
 L = Total length of wire

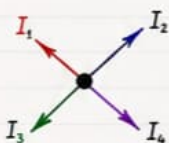
18 Kirchhoff's Laws

(i) Kirchhoff's Current Law (KCL)

At any junction, algebraic sum of currents is zero.

$$\sum I_{in} = \sum I_{out}$$

Based on conservation of charge.

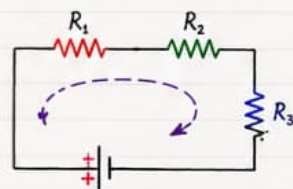


(ii) Kirchhoff's Voltage Law (KVL)

In any closed loop, algebraic sum of potential differences is zero.

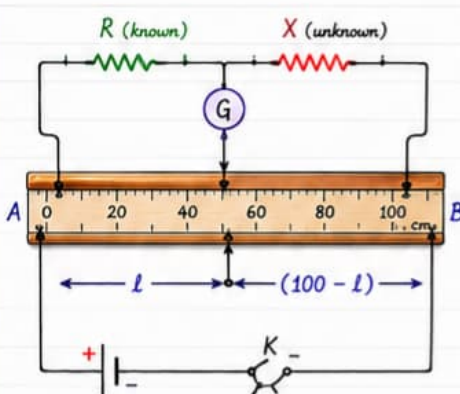
$$\sum V = 0$$

Based on conservation of energy.



19) Meter Bridge (Poggendorff's Method)

It is used to find the unknown resistance by comparing it with a known resistance.



Key Points

- Null point method is used.
- No current flows through galvanometer at balance point.

$$\frac{R}{X} = \frac{l}{100-l}$$

l = Balancing length
 R = Known resistance
 X = Unknown resistance

20) Dependence of Resistance

The resistance of a metallic conductor depends on its length, area of cross-section and temperature.

$$R = \rho \frac{l}{A}$$

Where,

ρ = Resistivity of material

l = Length

A = Area of cross-section

Effect of Temperature

$$R_T = R_0 [1 + \alpha (T - T_0)]$$

R_0 = Resistance at reference temperature T_0

R_T = Resistance at temperature T

α = Temperature coefficient of resistance ($^{\circ}\text{C}^{-1}$)

Key Points

- For metals, α is positive.
- For semiconductors and carbon, α is negative.

Typical Values of α (for some metals)

Metal	α ($^{\circ}\text{C}^{-1}$)	Metal	α ($^{\circ}\text{C}^{-1}$)
Silver	3.8×10^{-3}	Tungsten	4.5×10^{-3}
Copper	3.9×10^{-3}	Iron	5.0×10^{-3}
Aluminium	4.3×10^{-3}	Nichrome	0.4×10^{-3}

21) Applications of Current Electricity

- Electric heating (iron, geyser, heater)
- Electric lighting (LED, bulb)
- Electric motors and fans
- Electric power transmission
- Battery and cell based devices
- Measuring instruments
- Electroplating



Remember

Ohm's law is valid only for ohmic conductors and at constant physical conditions.

Common Units

- Current (I) $\rightarrow$ Ampere (A)
- Potential difference (V) $\rightarrow$ Volt (V)
- Resistance (R) $\rightarrow$ Ohm (Ω)
- Resistivity (ρ) $\rightarrow$ Ohm metre (Ωm)
- Power (P) $\rightarrow$ Watt (W)
- Energy (H) $\rightarrow$ Joule (J)

Quick Revision

- $I = \frac{dq}{dt}$
- $V = IR$
- $R = \rho \frac{l}{A}$
- $P = VI = I^2 R = \frac{V^2}{R}$
- $H = I^2 R t$
- $R_T = R_0 [1 + \alpha (T - T_0)]$
- $\sum I_{in} = \sum I_{out}$ (KCL)
- $\sum V = 0$ (KVL)

22) Solved Examples ★

Example 1:

A resistor of $10\ \Omega$ is connected to a $12\ \text{V}$ battery. Find the current.

Solution:

Given:

$$R = 10\ \Omega, V = 12\ \text{V}$$

Using Ohm's law,

$$I = \frac{V}{R} = \frac{12}{10} = 1.2\ \text{A}$$

Answer:

$$\text{Current} = 1.2\ \text{A}$$

Example 2:

Three resistors $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in parallel across a $12\ \text{V}$ battery. Find the current through each resistor and total current.

Solution:

In parallel, voltage across each resistor is same ($12\ \text{V}$).

$$I_1 = \frac{12}{2} = 6\ \text{A}$$

$$I_2 = \frac{12}{3} = 4\ \text{A}$$

$$I_3 = \frac{12}{6} = 2\ \text{A}$$

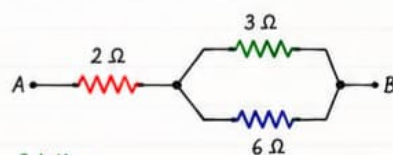
$$\text{Total current, } I = I_1 + I_2 + I_3 = 6 + 4 + 2 = 12\ \text{A}$$

Answer:

$$I_1 = 6\ \text{A}, I_2 = 4\ \text{A}, I_3 = 2\ \text{A}, I = 12\ \text{A}$$

Example 3:

Find the equivalent resistance of this network between A and B.



Solution:

Parallel part:

$$R_p = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\ \Omega$$

Total resistance,

$$R_{eq} = 2 + R_p = 2 + 2 = 4\ \Omega$$

Answer:

$$R_{eq} = 4\ \Omega$$